Lesson 02 Demo 01
Building a CrewAI-Powered Agent with OpenAI

Objective: To demonstrate how to build a multi-agent system using CrewAI that intelligently identifies queries requiring real-time information and retrieves the most up-to-date data accordingly

Scenario: A product manager at a digital knowledge platform needs to manage an increasing volume of user queries, many of which require real-time updates and dynamic information. At present, these queries are reviewed and answered manually, leading to delays, inconsistencies, and rising operational costs. The goal is to automate the query-handling process using a multi-agent system that can identify when real-time data retrieval is needed and provide users with fast, accurate, and context-aware responses while maintaining efficiency and cost control.

Tools required: Visual Studio Code

Prerequisites: Azure OpenAI key

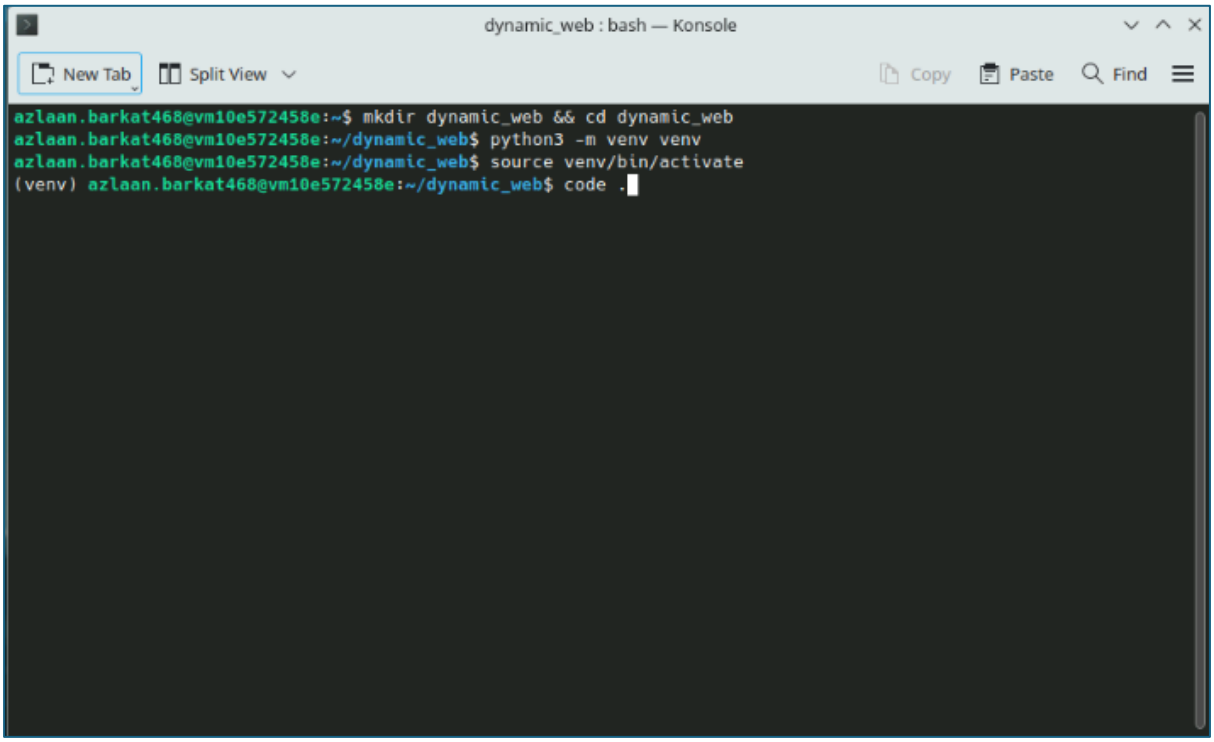
Steps to be followed:

1. Set up your local environment in Visual Studio Code
2. Configure the Tavily API Key and Azure OpenAI model
3. Define agents and tasks
4. Assemble and run the crew

Step 1: Set up your local environment in Visual Studio Code

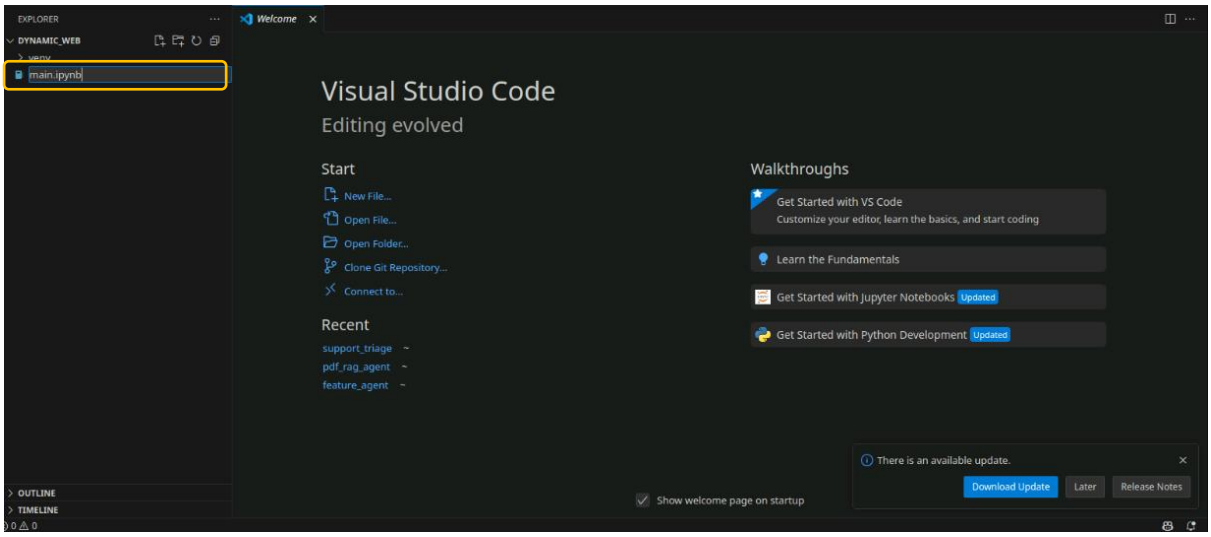
1.1 Open the terminal in your system and run the following commands:

mkdir dynamic_web && cd dynamic_web
python3 -m venv venv
source venv/bin/activate
code .



Note: This creates a virtual environment inside the dynamic_web folder and opens the project in VS Code.

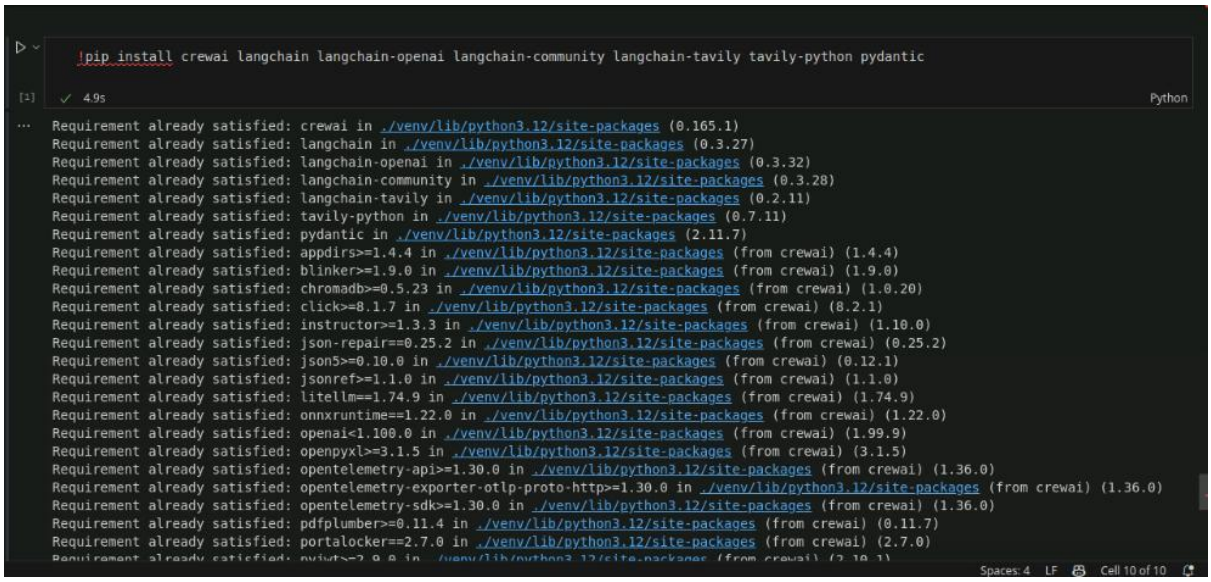
1.2 Create a new file and name it **main.ipynb**



1.3 In the first 2 cells, write and run the following commands to install the required packages:

!pip install crewai langchain langchain-openai langchain-community langchain-tavily tavily-python pydantic

!pip install "crewai[azure-ai-inference]"



```
main.ipynb > # ----------
Generate  + Code  + Markdown | Run All  Restart  Clear All Outputs | Jupyter Variables  Outline  ...  venv (Python 3.12.3)

!pip install "crewai[azure-ai-inference]"
[6]  ✓ 5.4s                                                                                    Python

...  Requirement already satisfied: crewai[azure-ai-inference] in ./venv/lib/python3.12/site-packages (1.10.1)
     Requirement already satisfied: aiosqlite~=0.21.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (0.21.0)
     Requirement already satisfied: appdirs~=1.4.4 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.4.4)
     Requirement already satisfied: chromadb~=1.1.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.1.1)
     Requirement already satisfied: click~=8.1.7 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (8.1.8)
     Requirement already satisfied: httpx~=0.28.1 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (0.28.1)
     Requirement already satisfied: instructor>=1.3.3 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.14.5)
     Requirement already satisfied: json-repair~=0.25.2 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (0.25.3)
     Requirement already satisfied: json5~=0.10.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (0.10.0)
     Requirement already satisfied: jsonref~=1.1.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.1.0)
     Requirement already satisfied: lancedb>=0.29.2 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (0.29.2)
     Requirement already satisfied: mcp~=1.26.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.26.0)
     Requirement already satisfied: openai<3,>=1.83.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (2.26.0)
     Requirement already satisfied: openpyxl~=3.1.5 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (3.1.5)
     Requirement already satisfied: opentelemetry-api~=1.34.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.34.1)
     Requirement already satisfied: opentelemetry-exporter-otlp-proto-http~=1.34.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-infe
     Requirement already satisfied: opentelemetry-sdk~=1.34.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.34.1)
     Requirement already satisfied: pdfplumber~=0.11.4 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (0.11.9)
     Requirement already satisfied: portalocker~=2.7.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (2.7.0)
     Requirement already satisfied: pydantic-settings~=2.10.1 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (2.10.1)
     Requirement already satisfied: pydantic~=2.11.9 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (2.11.10)
     Requirement already satisfied: pyjwt<3,>=2.9.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (2.11.0)
     Requirement already satisfied: python-dotenv~=1.1.1 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (1.1.1)
     Requirement already satisfied: regex~=2026.1.15 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (2026.1.15)
     Requirement already satisfied: textual>=7.5.0 in ./venv/lib/python3.12/site-packages (from crewai[azure-ai-inference]) (8.1.1)
     ...
     Using cached azure_core-1.38.2-py3-none-any.whl (217 kB)
                                                                                    {}  Cell 5 of 6
```

1.4 In the next cell, enter the following command to install litellm:

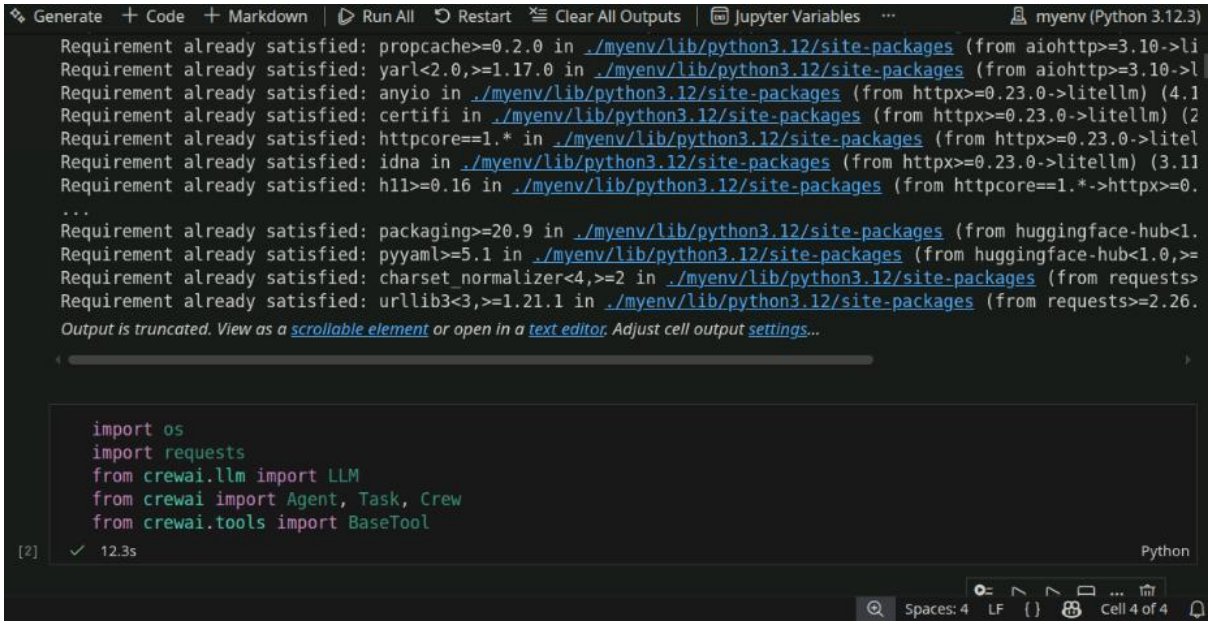
!pip install litellm

```
!pip install litellm
[1]  ✓ 2.5s                                                                                    Python

...  Requirement already satisfied: litellm in ./myenv/lib/python3.12/site-packages (1.82.0)
     Requirement already satisfied: aiohttp>=3.10 in ./myenv/lib/python3.12/site-packages (from litellm) (3.13.3)
     Requirement already satisfied: click in ./myenv/lib/python3.12/site-packages (from litellm) (8.1.8)
     Requirement already satisfied: fastuuid>=0.13.0 in ./myenv/lib/python3.12/site-packages (from litellm) (0.14.0)
     Requirement already satisfied: httpx>=0.23.0 in ./myenv/lib/python3.12/site-packages (from litellm) (0.28.1)
     Requirement already satisfied: importlib-metadata>=6.8.0 in ./myenv/lib/python3.12/site-packages (from litellm)
     Requirement already satisfied: jinja2<4.0.0,>=3.1.2 in ./myenv/lib/python3.12/site-packages (from litellm) (3.1
     Requirement already satisfied: jsonschema<5.0.0,>=4.23.0 in ./myenv/lib/python3.12/site-packages (from litellm)
     Requirement already satisfied: openai>=2.8.0 in ./myenv/lib/python3.12/site-packages (from litellm) (2.26.0)
     Requirement already satisfied: pydantic<3.0.0,>=2.5.0 in ./myenv/lib/python3.12/site-packages (from litellm) (2
     Requirement already satisfied: python-dotenv>=0.2.0 in ./myenv/lib/python3.12/site-packages (from litellm) (1.1
     Requirement already satisfied: tiktoken>=0.7.0 in ./myenv/lib/python3.12/site-packages (from litellm) (0.12.0)
     Requirement already satisfied: tokenizers in ./myenv/lib/python3.12/site-packages (from litellm) (0.20.3)
     Requirement already satisfied: aiohappyeyeballs>=2.5.0 in ./myenv/lib/python3.12/site-packages (from aiohttp>=3
                                                                                    Ln 1, Col 107 (106 selected)  Spaces: 4  Spaces: 4  LF  Cell 3 of 3
```

1.5 Import the required libraries using the following code:

import os
import requests
from crewai.llm import LLM
from crewai import Agent, Task, Crew
from crewai.tools import BaseTool



Step 2: Configure the Tavily API key and Azure OpenAI model

2.1 Use the Tavily API key to build a web search using the following code:

```
# Tavily API Key
TAVILY_API_KEY = "tvly-dev-1uT9rc-
X1i2A54uzUo18Hhz6BL1UjY8KMsU5aSCBbu3VRbNkr"

# ---- Custom CrewAI Tool for Web Search ----
class TavilySearchTool(BaseTool):
    name: str = "Web Search"
    description: str = "Search the web for recent information."

    def _run(self, query: str):
        url = "https://api.tavily.com/search"

        payload = {
            "api_key": TAVILY_API_KEY,
            "query": query,
            "max_results": 5
        }

        response = requests.post(url, json=payload)
        data = response.json()

        results = []
        for r in data["results"]:
```

```
        results.append(f"{r['title']} - {r['url']}")

    return "\n".join(results)


search_tool = TavilySearchTool()
```

```python
# Tavily API Key
TAVILY_API_KEY = "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"

# ---- Custom CrewAI Tool for Web Search ----
class TavilySearchTool(BaseTool):
    name: str = "Web Search"
    description: str = "Search the web for recent information."

    def _run(self, query: str):
        url = "https://api.tavily.com/search"

        payload = {
            "api_key": TAVILY_API_KEY,
            "query": query,
            "max_results": 5
        }

        response = requests.post(url, json=payload)
        data = response.json()

        results = []
        for r in data["results"]:
            results.append(f"{r['title']} - {r['url']}")

        return "\n".join(results)


search_tool = TavilySearchTool()
```

[3] ✓ 0.0s Python

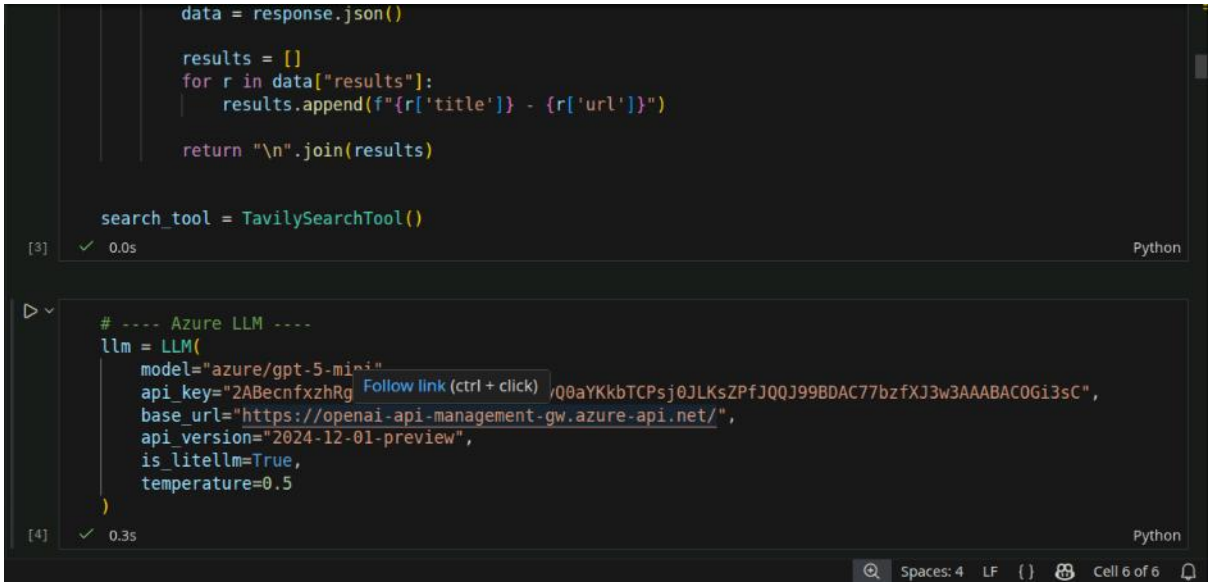
Spaces: 4 LF { } Cell 4 of 5

Note: Please create your own Tavily key and use it in the above code block.

2.2 Use the Azure OpenAI API key to build an LLM using the following code:

```
# ---- Azure LLM ----
llm = LLM(
   model="azure/gpt-5-mini",

api_key="2ABecnfxzhRg4M5D6pBKiqxXVhmGB2WvQ0aYKkbTCPsj0JLKsZPfJQQJ99
BDAC77bzfXJ3w3AAABACOGi3sC",
   base_url="https://openai-api-management-gw.azure-api.net/",
   api_version="2024-12-01-preview",
   is_litellm=True,
   temperature=0.5
)
```



Step 3: Define agents and tasks

3.1 Use the following code to create agents:

```
# ---- Agents ----
researcher = Agent(
   role="AI Researcher",
   goal="Find the latest advancements in AI for FMCG using web search",
   backstory="Expert in AI research trends who gathers real-time information from
the web.",
   verbose=True,
   llm=llm,
   tools=[search_tool]
)

writer = Agent(
   role="Technical Writer",
   goal="Summarize research into an executive report",
   backstory="Expert in executive summaries.",
   verbose=True,
   llm=llm
)
```

```python
# ---- Agents ----
researcher = Agent(
    role="AI Researcher",
    goal="Find the latest advancements in AI for FMCG using web search",
    backstory="Expert in AI research trends who gathers real-time information from the web.",
    verbose=True,
    llm=llm,
    tools=[search_tool]
)

writer = Agent(
    role="Technical Writer",
    goal="Summarize research into an executive report",
    backstory="Expert in executive summaries.",
    verbose=True,
    llm=llm
)
```

3.2 Define the research and writing tasks:

```python
# ---- Tasks ----
task_research = Task(
   description="Search the web and identify the top 3 recent advancements in AI for FMCG.",
   expected_output="Detailed notes explaining three recent AI advancements in FMCG with examples.",
   agent=researcher
)

task_write = Task(
   description="""
Write a concise executive summary using the research notes.

Requirements:
- Maximum 200 words
- Use bullet points
- Focus only on the 3 key advancements
""",
   expected_output="Executive summary of AI advancements in FMCG.",
   agent=writer,
   context=[task_research]
)
```

```python
# ---- Tasks ----
task_research = Task(
    description="Search the web and identify the top 3 recent advancements in AI for FMCG.",
    expected_output="Detailed notes explaining three recent AI advancements in FMCG with examples.",
    agent=researcher
)

task_write = Task(
    description="""
Write a concise executive summary using the research notes.

Requirements:
- Maximum 200 words
- Use bullet points
- Focus only on the 3 key advancements
""",
    expected_output="Executive summary of AI advancements in FMCG.",
    agent=writer,
    context=[task_research]
)
```

Step 4: Assemble and run the crew

4.1 Use the code given below to arrange all the components of the crew and run it:

```python
# ---- Crew ----
crew = Crew(
    agents=[researcher, writer],
    tasks=[task_research, task_write],
    verbose=True
)

result = crew.kickoff()

print("\nFinal Output:\n")
print(result.raw)
```

```python
# ---- Crew ----
crew = Crew(
    agents=[researcher, writer],
    tasks=[task_research, task_write],
    verbose=True
)

result = crew.kickoff()

print("\nFinal Output:\n")
print(result.raw)
```

The program starts generating a summary report, as shown in the following screenshots:

```
Crew Execution Started

Crew Execution Started
Name:
crew
ID:
bf947ad6-f66f-47c0-9b00-d9ac1c94cb5d

Task Started

Task Started
Name: Search the web and identify the top 3 recent advancements in AI for FMCG.
ID: d982c1b2-72b5-42be-8f18-550b10bdad05

Ln 20, Col 2   Spaces: 4   LF   { }   Cell 8 of 8
```

```
Agent Final Answer

Agent: AI Researcher

Final Answer:
Top 3 recent AI advancements in FMCG (detailed notes with examples)

1) AI-powered probabilistic & causal demand forecasting (next-gen forecasting for planning and supply)
- What's changed recently
  - Shift from point-forecast ML to probabilistic forecasting (models that output full demand distributions
and prediction intervals rather than single-point forecasts). This enables better risk-aware decisions
(inventory buffers, service-level tradeoffs) and scenario planning.
  - Increased use of causal and hybrid models that combine time-series ML with causal drivers (promotions,
macro/market signals, weather, competitor behavior) to isolate effects and support "what-if" simulations.
  - Faster model cycles and MLOps/augmentation so forecasting models are refreshed more frequently and tiec
directly into S&OP/IBP systems (shorter lead times from data to actionable plan).
  - Adoption of external alternative data (mobility, search trends, social, syndicated POS) and transfer
learning across similar SKUs/regions to improve cold-start forecasts for new SKUs/markets.
- Why it matters for FMCG
  - FMCG margins and service levels are highly sensitive to stockouts and excess inventory; probabilistic
forecasts allow optimization that trades off waste, storage costs and lost sales in a principled way.
  - Enables more robust promotions planning and improved allocation in omni-channel environments.
- Vendor & real-world examples
  - Blue Yonder / o9 Solutions / other supply-chain AI vendors: recent product updates and case studies
(2024–25) emphasize probabilistic forecasting, scenario planning, and embedding ML models into planning
workflows. (See vendor release notes and o9 case studies describing demand planning deployments for food &
beverage customers.)
  - Large FMCG players (e.g., Nestlé, Coca-Cola, major retailers) are documented moving to ML-based demand

Ln 20, Col 2   Spaces: 4   LF   { }   Cell 8 of 8
```

```
...
    Final Output:

    - Probabilistic & causal demand forecasting — next-gen planning: recent shift from point forecasts to probabili

    - Generative AI for creative production, personalization & campaign automation: mature text/image/video models

    - Computer vision & shelf intelligence for real-time in-store execution: edge CV and improved sensing deliver r
...
                                    Crew Completion

    Crew Execution Completed
    Name:
    crew
    ID:
    bf947ad6-f66f-47c0-9b00-d9ac1c94cb5d
    Final Output: - Probabilistic & causal demand forecasting — next-gen planning: recent shift from point
    forecasts to probabilistic and causal/hybrid models (with alternative data and faster MLOps) enables
    scenario-based, risk-aware inventory and promotions planning; delivers lower forecast error, improved
    safety-stock sizing and faster S&OP turnaround; in production with vendors like Blue Yonder/o9 and major
    FMCG—requires strong data integration, governance and planner adoption.
```

By following these steps, you have successfully built a CrewAI-powered multi-agent system that intelligently routes and processes user queries, determines when real-time web data is needed, and retrieves the latest results via the Tavily Search Tool.